Enoch Larrey, **Ph.D.**

Bioinformatics Research Scientist | Computational Biologist | Multi-Omics Data Scientist

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PROFESSIONAL SUMMARY

Bioinformatics research scientist with a Ph.D. in Bioinformatics and 4+ years of experience developing reproducible analytical workflows for bulk RNA-seq, single-cell RNA-seq, quantitative proteomics, microbial genomics, machine learning, and integrated multi-omics studies. Proficient in R, Python, Bash, Linux/Unix, high-performance computing, Bioconductor, Seurat, DESeq2, limma, and publication-quality visualization. Experienced partnering with experimental scientists to design analyses, interpret complex datasets, mentor junior researchers, document computational methods, and translate findings into manuscripts, conference presentations, and biologically actionable insights. Research background spans cancer biology, immunology, endothelial biology, radiation injury, and organoid systems, with strong interest in applying computational approaches to neuroscience, neuro-oncology, and pediatric brain tumor research.

CORE CAPABILITIES

- Development, optimization, and documentation of reproducible bioinformatics pipelines for high-dimensional sequencing and proteomics datasets.
- Single-cell RNA-seq analysis using Seurat & Scanpy: quality control, normalization, dimensionality reduction, clustering, cell-type annotation, differential expression, and biological interpretation.
- Bulk RNA-seq and quantitative proteomics analysis: preprocessing, statistical modeling, differential expression/abundance, PCA, heatmaps, volcano plots, pathway enrichment, and interaction-network analysis.
- Statistical computing and machine learning using R and Python, including regression, classification, feature importance, clustering, and model evaluation.
- Familiar with analytical principles underlying GWAS enrichment, network biology, and spatial omics analyses through independent study and related computational experience, with a strong interest in expanding expertise in spatial transcriptomics, spatial proteomics, and neuronal activity analysis.
- Cross-functional collaboration, research coordination, mentoring, scientific writing, data visualization, and rigorous computational documentation.

TECHNICAL SKILLS

- **Programming & computing:** R, Python, Bash, Linux/Unix, HPC, Git, GitHub, Jupyter, RStudio
- **Transcriptomics & single-cell:** RNA-seq, scRNA-seq, Seurat, DESeq2, limma, Bioconductor, STAR, BWA, GATK, FastQC, Trimmomatic, TopHat
- **Proteomics:** Spectronaut, Scaffold DIA, proteoDA, quality assessment, normalization, differential abundance, Ingenuity Pathway Analysis, pathway enrichment, protein interaction networks
- **Statistics & machine learning:** Pandas, NumPy, scikit-learn, TensorFlow, XGBoost, SVM, Random Forest, logistic regression, KNN, gradient boosting, AdaBoost, hierarchical clustering, survival analysis
- **Visualization & reporting:** ggplot2, ComplexHeatmap, matplotlib, seaborn, Power BI, GraphPad Prism, Minitab; PCA, UMAP/t-SNE, volcano plots, heatmaps, Kaplan-Meier curves
- **Genomics & microbial analysis:** Whole-genome analysis, SPAdes, Pilon, Prokka, QUAST, Abricate, Sickle, variant analysis, antimicrobial-resistance screening

PROFESSIONAL EXPERIENCE

Doctoral Researcher, Bioinformatics

University of Arkansas for Medical Sciences | Little Rock, AR | Aug 2021- Jul 2026

- Developed and maintained reproducible R, Bash, and Bash workflows for processing and statistical analysis of transcriptomic, proteomic, and genomic datasets.
- Analyzed bulk RNA-seq and quantitative proteomics data to identify protein expression patterns, enriched pathways, and molecular networks associated with tissue injury, cancer biology, platelet function, and endothelial-cell responses.
- Integrated proteomic findings with RT-qPCR, Western blotting, immunohistochemistry, immunofluorescence, and organoid-model results to strengthen biological interpretation and experimental follow-up.
- Generated publication-quality figures and research-ready summaries, including PCA plots, heatmaps, volcano plots, enrichment visualizations, and protein-interaction networks for manuscripts, presentations, and multidisciplinary project discussions.
- Collaborated closely with experimental scientists on study design, sample metadata, analysis strategy, interpretation, and next-step prioritization across multiple concurrent research projects.

- Mentored summer research students in experimental design, data analysis, interpretation, and scientific presentation; supported research progress through regular guidance and troubleshooting.
- Contributed to seven peer-reviewed publications, first-author manuscripts, conference abstracts, posters, and an oral presentation at national and international conferences.

Graduate Research Fellow, Bioinformatics

COMSATS University Islamabad | Islamabad | Pakistan | 2018 – 2020

- Developed a reverse-vaccinology workflow to screen *Mycobacterium africanum* proteome, identify essential and non-homologous proteins to human proteins, predict immunogenic B- and T-cell epitopes, and prioritize vaccine candidates.
- Performed protein-structure prediction, quality assessment, molecular docking, and computational interpretation using MODELLER, AutoDock Vina, UCSF Chimera, and PyMOL.
- Maintained reproducible analysis records and prepared figures, tables, and written summaries for thesis and publication activities.

SELECTED PROJECTS

- **Single-Cell RNA-Seq Analysis of Non-Small Cell Lung Cancer** Project Link**:** Analyzed 10x Genomics data using Seurat; performed QC, normalization, scaling, clustering, cell-type annotation, differential expression, and characterization of tumor-microenvironment heterogeneity.
- **Radiation-Induced Intestinal Injury Multi-Omics Analysis** Project Link**:** Analyzed RNA-seq and quantitative proteomics data to identify genes, proteins, pathways, and candidate biomarkers associated with intestinal radiation toxicity and epithelial injury repair.
- **Microbial Whole-Genome Analysis** Project Link**:** Processed raw bacterial whole-genome sequencing data, performed quality control, de novo assembly, annotation, antimicrobial-resistance screening, variant analysis, and phylogenetic interpretation to assess the pathogenicity of the bacteria present in food.
- **Machine Learning for Disease Classification** Project Link**:** Built and compared SVM, Random Forest, XGBoost, CNN, and RNN models for chronic kidney disease and breast-cancer classification; evaluated performance with classification metrics and feature importance to facilitate early detection of these diseases for rapid treatment.

EDUCATION

Ph.D. Bioinformatics — **July 2026**
University of Arkansas at Little Rock & University of Arkansas for Medical Sciences — Little Rock, AR

M.Sc. Bioinformatics — **July 2020**
COMSATS University Islamabad — Islamabad, Pakistan

B.Sc. Biotechnology and Molecular Biology — **July 2017**
University for Development Studies — Tamale, GH

PEER-REVIEWED PUBLICATIONS

- Nounamo B, Cholia RP, Du R, **Larrey EK**, Pineda EN, Porter R, Ware J, Pathak R. ***Identification and Characterization of Platelet Surface-membrane-derived Microparticles before and after Total-body Irradiation***. Radiat Res. 2025 Dec 12. doi: 10.1667/RADE-25-00165.1. Epub ahead of print. PMID: 41421387.
- McGinness K, Al-Anbaky Q, **Larrey EK**, Cholia RP, Rotenberry M, Cook H, Pineda EN, Du R, Boerma M, Pathak R. ***Geranylgeranyl Transferase Inhibitors Mitigate Intestinal Radiation Toxicity***. Int J Radiat Oncol Biol Phys. 2025 Nov 4:S0360-3016(25)06418-1. doi: 10.1016/j.ijrobp.2025.10.026. Epub ahead of print. PMID: 41197739; PMCID: PMC12742174.
- Pineda EN, Nounamo B, Du R**, Larrey EK**, Gilreath C, Cook H, Boerma M, Koturbash I, Pathak R. ***Sex-specific immune alterations in mice following long-term simulated microgravity and chronic irradiation.*** NPJ Microgravity. 2025 Jun 12;11(1):24. doi: 10.1038/s41526-025-00480-1. PMID: 40506441; PMCID: PMC12162859.
- Reyes Y, **Larrey EK**, Pathak R, Veisaga ML, Barbieri MA, Ward S, Kumar A, Sevilla MD, Adhikary A, Wnuk SF. ***Azido derivatives of sesquiterpene lactones: Synthesis, anticancer proliferation, and chemistry of nitrogen-centered radicals***. Results Chem. 2024 Jul;9:101643. doi: 10.1016/j.rechem.2024.101643. Epub 2024 Jul 5. PMID: 39498431; PMCID: PMC11533910.
- **Larrey EK**, Pathak R. ***Radiation-Induced Intestinal Normal Tissue Toxicity: Implications for Altered Proteome Profile***. Genes. 2022; 13(11):2006. https://doi.org/10.3390/genes13112006
- Ohwofasa, A., Aliyu, A. A., **Larrey EK**, & Ogbodu, O. D. (2020). ***Characterization of crude invertase extracted from Aspergillus species CSA 35***. African Journal of Biotechnology , 19(8), 584-590
- **Larrey EK**., Laryea, JNA., Kpordze, SW., & Saba, CKS. (2020). ***Microbial load of indoor airborne bacteria and fungi in a teaching hospital in Ghana***. African Journal of Microbiology Research, 14(3), 100-105

PROFESSIONAL MEETINGS/CONFERENCES/PRESENTATIONS

- **Larrey EK,** Pathak R. (2025). ***Platelet GPIbα dysfunction suppresses intestinal organoid growth prior to and***

following total body irradiation in mice. Oral presentation at Radiation Research Society, San Juan, Puerto Rico

- **Larrey EK**, Harrison Cook H, Garg S, Al-anbaky Q, Cholia RQ, Nounamo BC, Binz R, Pathak R. (2024). ***KLF2 Haploinsufficiency Exacerbates Intestinal Radiation Damage by Disrupting Stromal-Epithelial Crosstalk***. Poster session presented at Radiation Research Society, Tucson, AZ, USA
- **Larrey EK**, Sotomayor ENP, Cholia RP, Al-anbaky Q, Nounamo BC, and Pathak R. (2024). ***Implication of Tumor Suppressor Gene KLF2 on Mouse Intestinal Organoid Formation***. Poster session presented at Winthrop P. Rockefeller Cancer Institute Retreat, Little Rock, AR, USA
- **Larrey EK**, Byrum SD, Ware J, Boerma M, Pathak R. (2023). ***Compared to wild-type mice, mice deficient of platelet glycoprotein-Ibα (GPIbα) or -VI (GPVI) exhibit altered microparticle generation, inflammation, and platelet protein profile prior to or following irradiation***. Poster session presented at International Congress for Radiation Research, Montreal, MTL, Canada
- **Larrey EK**, Byrum SD, Pathak R. (2023). ***Quantitative proteomics: Modulation of molecular pathways involved in cancer progression by the overexpression of somatostatin receptor-2 or -3***. Poster session presented at the 19th Annual Meeting of MCBIOS, MidSouth Computational Biology and Bioinformatics Society, Dallas, TX, USA
- **Larrey EK**, Yahaira R, Amitava A, Stanislaw FW, Pathak R. (2023). ***Synergistic suppression of human breast cancer cells by combination of gamma tocotrienol and azido-parthenolid***. Poster session presented at Winthrop P. Rockefeller Cancer Institute Retreat, Little Rock, AR, USA

SELECTED GRADUATE LEVEL COURSES

Data Science, Statistics & Biostatistics: Artificial Intelligence, Introduction to R, Biostatistics for Bioinformatics, Biostatistics I, Information Visualization, Biological Data Analysis
Computational Biology, Bioinformatics & Drug Discovery: Bioinformatics Theory and Application, Introduction to Bioinformatics, Computer Aided Drug Design, Molecular modeling
Biology: Advances in Molecular Biology, Molecular Biology I, Biochemistry and Metabolism, Cellular and Molecular Biology, Systems Biology of Microorganisms.

COMPLETED TRAININGS

- Machine Learning, Data Science and Generative AI with Python (UDEMY)**:** **Credential ID**
- Getting Started with Deep Learning (NVIDIA)**:** **Credential ID**
- SQL Bootcamp: Go from Zero to Hero
- NCGR 2024 Virtual Bioinformatics Workshop on Pangenomics (NM-INBRE and NCGR)**:** **Credential ID**
- Arkansas Summer Research Institute Certificate of Completion**:** **Credential ID**
- Genomics Workshop (ArC-GEM)**:** **Credential ID**

PROFESSIONAL DEVELOPMENT – IN PROGRESS

- AWS Certified Machine Learning Engineer Associate - Exam Preparation, In Progress
- AWS Certified AI Practitioner - Exam Preparation, In Progress
- AWS Certified Cloud Practitioner - Exam Preparation, In Progress
- AI Automation: Build LLM Apps & AI-Agents with n8n & APIs

AWARDS AND HONORS

- **Scholar in Training Travel Awards,** Radiation Research Society **2024 & 2025**
- **Outstanding Doctoral Research Award**, UALR/UAMS Graduate Program in Bioinformatics **05/2023**
  Certificate for Outstanding academic performance and commitment to excellence in research.
- **Graduate Research Assistantship,** UALR/UAMS **2021 & 2026**
- **Queen Elizabeth Commonwealth Scholarship,** Association of Commonwealth Universities **2018-2020**

PROFESSIONAL MEMBERSHIPS/AFFILIATIONS

- **Member, Midsouth Computational Biology and Bioinformatics Society (MCBIOS)** January 2022 – July 2026
- **Member, Radiation Research Society** January 2022 – July 2026